Pricing an S&P 500 Autocallable Contingent Coupon Note with a Calibrated Heston Model

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Executive Summary

Objective: Value a Bank of America four-year autocallable contingent coupon note on the S&P 500 and test whether a stochastic volatility model changes the answer compared with a standard flat-volatility model.

Approach: I calibrated a Heston stochastic volatility model to the SPX implied volatility surface (Bloomberg, fit error 0.19 vol points RMSE) and priced the note by Monte Carlo simulation, using Bloomberg rates and dividends as of the pricing date ($r = 4.57\%$, $q = 1.15\%$). As a benchmark, I priced the same note with a flat-volatility binomial tree and a flat-volatility Monte Carlo.

Results:

- **Fair value: $998.56 per $1,000** (99% confidence interval $995.65–$1,001.47). This is **0.56% above** BofA's estimated value of $993.00, a gap consistent with an issuer funding spread of **0.40%** over the risk-free rate from the USD SOFR overnight index swap (OIS) curve (Bloomberg curve S490).
- **The numbers are robust:** 20,000 paths keep the probability of a pricing error above $10 (1% of principal) below 1.3%; the price is stable to the simulation time step; and the flat-volatility tree and Monte Carlo agree within simulation error.
- **Flat volatility gets the price but not the risk:** At the at-the-money volatility (20.89%), a flat-volatility model prices the note within $2 of Heston, but for different reasons. Heston gives a 68% chance of early redemption on the first call date versus 52%, a shorter expected life (1.6 vs 1.9 years), and a lower probability of losing principal (7.4% vs 11.1%) with a much deeper loss tail. Using the barrier-level volatility (25.40%) instead underprices the note by about $31.

Takeaway: A flat-volatility model can reproduce the price only if the right volatility is chosen, and even then it misstates the timing of early redemption and the size of potential losses, which drive how the note should be hedged.

1 The Product

The note is a four-year **autocallable contingent coupon note** issued by Bank of America on the S&P 500 Index, with $1,000 principal (terms from the issuer's pricing supplement, SEC Form 424B2). Its payoff depends on the index level on 16 quarterly observation dates, measured against the initial level $S_0 = 7{,}650.50$.

- **Contingent coupon feature:** On each observation date, if the index closes at or above 70% of S_0, the note pays a coupon of 2% of principal ($20, or 8% per year). Otherwise no coupon is paid for that quarter.
- **Automatic call feature:** From the fourth observation date (20-Sep-2027) onward, if the index closes at or above S_0, the note is redeemed early: the investor receives the $1,000 principal plus that quarter's coupon, and the note terminates.
- **Maturity payoff:** If the note is not called before maturity, the investor receives the full principal if the final index level is at or above 70% of S_0. If it is below, the investor receives $P \cdot S_T/S_0$ and loses principal one-for-one with the index decline.
- **Payment dates:** Each payment settles 3 business days after its observation date.

Why a model choice matters: The note behaves like a bond that is likely to be repaid early, combined with a short position in a downside put on the index. Its value depends on the whole path of the index, not just its final level: on whether and *when* the index first closes above S_0, which ends the note, and on how far it can fall below 70% if it is never called. Both depend on how volatility behaves over time and across index levels, which is why the note is priced with a stochastic volatility model rather than a single flat volatility.

Term	Notation	Value
Issuer	–	Bank of America
Underlying	–	S&P 500 Index
Pricing date	–	18-Sep-2026
Final observation date	–	18-Sep-2030
Maturity date	–	23-Sep-2030
Principal	P	\$1,000
Initial index level	S_0	7,650.50
Autocall level	L_{call}	100% of S_0
Coupon barrier	L_{coupon}	70% of S_0
Coupon per quarter	c	2% of principal (8% per year)
Downside threshold	$L_{\text{threshold}}$	70% of S_0
Observation dates	–	16, quarterly
Autocall dates	–	12, from 20-Sep-2027 to 18-Jun-2030
Issuer estimated value	–	\$993.00

Table 1. Contract terms.

2 Market Data

All market data come from a Bloomberg terminal (Margolis Market Information Lab, University of Illinois Urbana-Champaign). Table 2 lists the inputs used in the pricing model.

Input	Value	Source
Initial index level S_0	7,650.50	BofA term sheet, 18-Sep-2026
Risk-free rate r	4.567% (continuous)	USD SOFR OIS curve, from 18-Sep-2026 to 23-Sep-2030
Dividend yield q	1.151% (continuous)	Bloomberg OVME, from 18-Sep-2026 to 23-Sep-2030
Implied volatility surface	8 expiries, 200 options	Bloomberg OMON, 12-Oct-2026
ATM volatility term structure	19 expiries	Bloomberg OVDV, 12-Oct-2026
Issuer estimated value	\$993.00	BofA term sheet, 18-Sep-2026

Table 2. Market and contract inputs.

Rate and dividend: The risk-free rate is the four-year zero rate from the USD SOFR overnight index swap (OIS) curve, the market's standard risk-free discount curve; any premium the issuer pays for its own credit is therefore measured separately, as a funding spread over OIS (Section 4). The dividend yield is Bloomberg's continuous dividend yield over the same horizon from the OVME option pricer. Both are taken on the pricing date and held constant over the life of the note. Together they reproduce Bloomberg's four-year forward of 8,788.93 to within day-count conventions.

Volatility data: The model is calibrated to the *live* SPX implied volatility surface on 12-Oct-2026 (15:00 CDT, spot 7,798.91) rather than to a historical surface for the 18-Sep-2026 pricing date, because historical option surfaces on the terminal were of poor quality. Three datasets are used, one for each calibration step:

1. **ATM term structure:** At-the-money-forward implied volatility for the 19 standard monthly (third-Friday) expiries from 16-Oct-2026 to 20-Dec-2030, the most liquid SPX contracts.
2. **Short-dated smiles:** Full option chains for the 18-Dec-2026 and 19-Mar-2027 expiries.
3. **Volatility surface:** Eight expiries from 19-Mar-2027 to 20-Dec-2030, 25 strikes each, covering forward moneyness K/F from about 0.70 to 1.03, the range that matters for the note's 70% and 100% levels.

Cleaning procedure: Quotes with a zero bid or fewer than 10 contracts traded are removed, and only out-of-the-money options are kept (puts below the forward, calls above it). In the short-dated smiles, puts below 60% of the forward are also removed: their prices are only a few ticks, which makes their implied volatility spreads unusually wide. Implied volatilities are backed out from bid, ask and mid prices with Black's formula, using Bloomberg's forward for each expiry and a discount factor implied by put-call parity. The bid-ask spread in implied volatility is kept as a measure of quote quality.

3 Model and Calibration

3.1 The Heston model

Under the risk-neutral measure, the index and its variance follow

$$\begin{aligned} dS_t &= (r-q)\,S_t\,dt + \sqrt{v_t}\,S_t\,dW_t^S, \\ dv_t &= \kappa(\theta - v_t)\,dt + \eta\sqrt{v_t}\,dW_t^v, \qquad d\langle W^S, W^v\rangle_t = \rho\,dt, \end{aligned} \tag{1}$$

where v_0 is the current variance, θ the long-run variance, κ the speed of mean reversion, η the volatility of variance and ρ the correlation between index and variance shocks. Heston produces the two features that matter for this note: a **term structure** of volatility ($v_0 \neq \theta$) and a **skew**, with higher implied volatility for low strikes ($\rho < 0$). European option prices have a semi-closed form through the characteristic function.

3.2 Calibration

The five parameters are fitted in three steps. The first two give a reasonable initial guess for the final five-parameter fit. The full optimization problem and the option pricing formula are given in Appendices A and B.

Step 1: v_0, κ, θ from the ATM term structure. Under Heston, the expected average variance to maturity T is $\bar{v}(T;\kappa,\theta,v_0) = \theta + (v_0-\theta)\,\dfrac{1-e^{-\kappa T}}{\kappa T}$. With v_0 set to the ATM variance of the nearest monthly expiry (16-Oct-2026), κ and θ are fitted to the market ATM implied variances $\sigma^2_{ATM}(T_i)$ of the 19 monthly expiries:

$$\kappa', \theta' = \arg\min_{\kappa,\theta} \sum_i \left[\bar{v}(T_i;\kappa,\theta,v_0) - \sigma^2_{ATM}(T_i)\right]^2. \tag{2}$$

Step 2: η, ρ from the short-dated smiles. With v_0, κ', θ' fixed, η and ρ are fitted to the December 2026 and March 2027 smiles:

$$\eta', \rho' = \arg\min_{\eta,\rho} \sum_i \left[\sigma^{Heston}_{imp,i} - \sigma^{market}_{imp,i}\right]^2. \tag{3}$$

Step 3: all five parameters on the full surface. Starting from steps 1–2, all parameters are fitted to the 200-option surface:

$$v_0^*, \kappa^*, \theta^*, \eta^*, \rho^* = \arg\min_{v_0,\kappa,\theta,\eta,\rho} \sum_{i=1}^{200} \left[\sigma^{Heston}_{imp,i} - \sigma^{market}_{imp,i}\right]^2. \tag{4}$$

Two alternative objectives were also tested: weighting each error by the option's bid-ask spread in implied volatility, $w_i = \max(0.0005,\ \sigma^{ask}_{imp,i} - \sigma^{bid}_{imp,i})$, and adding a penalty that enforces the Feller condition $2\kappa\theta \geq \eta^2$. All fits use bounded least squares.

3.3 Results

Fit	v_0	κ	θ	η	ρ	RMSE (vol pts)
Regular (selected)	**0.0283**	**3.416**	**0.0547**	**1.950**	**−0.545**	**0.19**
Spread-weighted	0.0191	4.203	0.0481	1.595	−0.604	0.28
Feller-penalized	0.0146	1.880	0.0473	0.422	−0.950	0.36

Table 3. Calibrated Heston parameters and fit error on the volatility surface.

I use the **regular fit**, which has the lowest error, with a root-mean-square error of 0.19 volatility points against the market (Figure 1). The parameters are economically sensible: a current volatility of $\sqrt{v_0} = 16.8\%$, rising toward a long-run level of $\sqrt{\theta} = 23.4\%$, with mean reversion ($\kappa = 3.4$, half-life about 2.5 months) and a negative correlation that produces the equity skew. The final fit is insensitive to the starting point: changing the step-2 inputs leads to the same parameters to about four digits.

Feller condition: The regular fit does not satisfy the Feller condition: $2\kappa\theta/\eta^2 = 0.098$, against the required ≥ 1. This is common for equity indices, where a large η is needed to match the steep short-dated skew. The condition only guarantees that the variance stays strictly positive. Imposing it nearly doubles the fit error (RMSE 0.19 to 0.36 vol points) and drives ρ to its bound of -0.95, a sign of underfitting. The simulation therefore uses the unconstrained

parameters and handles variance reaching zero with a full-truncation scheme (Section 4). The penalty weight was not tuned, so a milder constraint may fit better (Section 7.2).

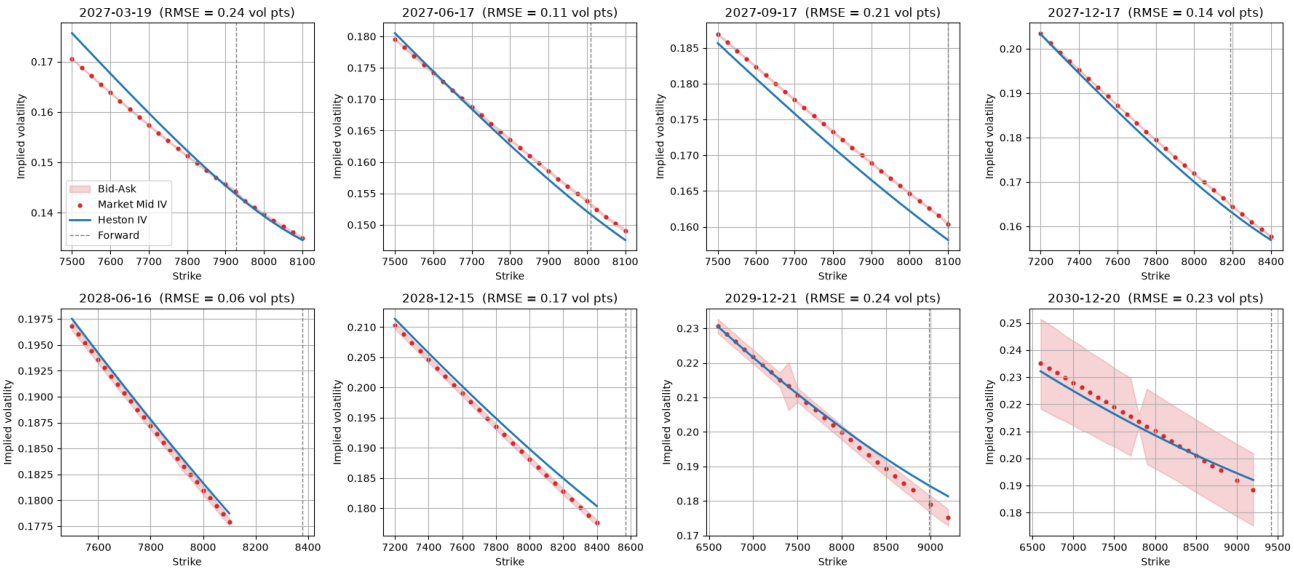


Figure 1. Heston implied volatility (regular fit) against market quotes and bid–ask band, by expiry.

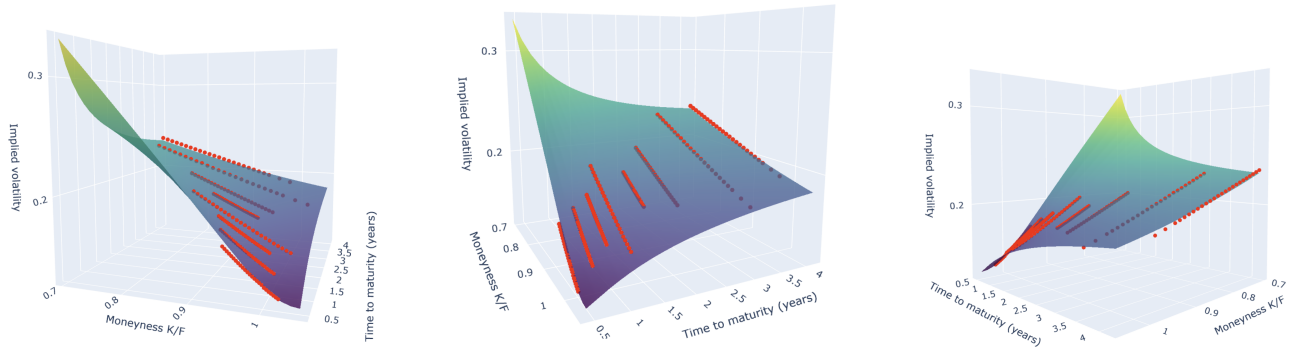


Figure 2. Calibrated Heston implied volatility surface from three angles, with market quotes (red).

4 Pricing

4.1 Monte Carlo setup

The index and variance are simulated on a daily grid from the pricing date to the final observation date ($N = 1{,}461$ steps, $\Delta t = 1/365$) for $M = 20{,}000$ paths. The index uses a log-Euler step and the variance a full-truncation Euler step, which floors the variance at zero inside the drift and diffusion so the stock process stays well defined when the Feller condition fails:

$$\begin{aligned} v_i^+ &= \max(v_i, 0), \\ S_{i+1} &= S_i \exp\left[\left(r - q - \tfrac{1}{2} v_i^+\right)\Delta t + \sqrt{v_i^+ \Delta t}\, Z_{1,i}\right], \\ v_{i+1} &= v_i + \kappa\left(\theta - v_i^+\right)\Delta t + \eta\sqrt{v_i^+ \Delta t}\left(\rho Z_{1,i} + \sqrt{1-\rho^2}\, Z_{2,i}\right), \end{aligned} \tag{5}$$

with $Z_{1,i}, Z_{2,i}$ independent standard normals. On each observation date the path is checked against the coupon barrier and, from the fourth observation date on, the autocall level; a called path stops. Every cash flow is discounted from its payment date at the OIS rate, and the price is the average discounted payoff over all paths. The full algorithm is given in Appendix C.

4.2 Result

	Value per \$1,000
Heston Monte Carlo fair value	**998.56**
99% confidence interval	[995.65, 1,001.47]
BofA estimated value	993.00
Difference (Heston − BofA)	+5.56 (+0.56%)
Implied issuer funding spread over OIS	0.40%

Table 4. Fair value of the note.

The fair value is **\$998.56**, just below par. BofA's estimated value of \$993.00 lies outside the 99% confidence interval, so **the \$5.56 difference is not simulation noise**.

Why the issuer's value is lower: The note is an unsecured obligation of Bank of America, so the issuer values it by discounting at its own *funding rate* rather than at the risk-free OIS rate used here. Solving for the spread over OIS that reproduces BofA's value gives **0.40%**, a realistic funding spread for a four-year senior bank note. The model's fair value is therefore consistent with the issuer's estimate once the issuer's credit is taken into account.

Sensitivity to the calibration choice: Repricing with the alternative calibrations of Section 3 gives \$1,000.63 (spread-weighted) and \$1,000.11 (Feller-penalized), within about \$2 of the selected fit and inside the Monte Carlo confidence interval. The price is therefore not sensitive to the choice of calibration objective.

5 Validation

Three checks confirm that the price does not depend on the numerical settings: the number of simulated paths, the time step, and an independent binomial tree. Details are in Appendix D.

5.1 Number of paths

With $M = 20{,}000$ paths, the standard deviation of the discounted payoff is $\hat{\sigma} = 159.7$. By the weak law of large numbers (Chebyshev's inequality), the probability that the Monte Carlo price misses the true model price by more than $\epsilon = \$10$ (1% of principal) is at most

$$P\left(|\hat{V}_M - V| \geq \epsilon\right) \leq \frac{\hat{\sigma}^2}{M\,\epsilon^2} = \frac{159.7^2}{20{,}000 \times 10^2} = 1.3\%. \tag{6}$$

This bound holds for any payoff distribution. The central limit theorem gives a tighter 99% confidence interval, $\hat{V}_M \pm 2.576\,\hat{\sigma}/\sqrt{M} = 998.56 \pm 2.91$. The payoff itself is strongly skewed (skewness -3.7, from the possibility of losing principal), but the skewness of the sample mean is only $-3.7/\sqrt{20{,}000} \approx -0.03$, so the normal approximation behind the interval is reliable (Figure 3).

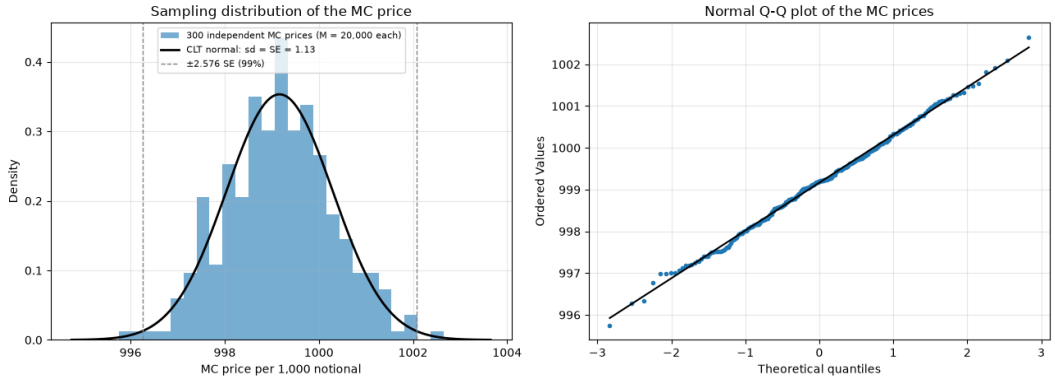


Figure 3. MC price over 300 independent runs: histogram against the CLT normal density (left) and normal Q–Q plot (right).

5.2 Time step

Repricing on uniform grids of $N = 1{,}000$ to 2,000 steps gives prices between \$996.97 and \$998.64 (Figure 4). The range of \$1.67 is smaller than the 99% half-width (about \$2.9), so the variation is simulation noise, and the daily grid ($N = 1{,}461$) is fine enough.

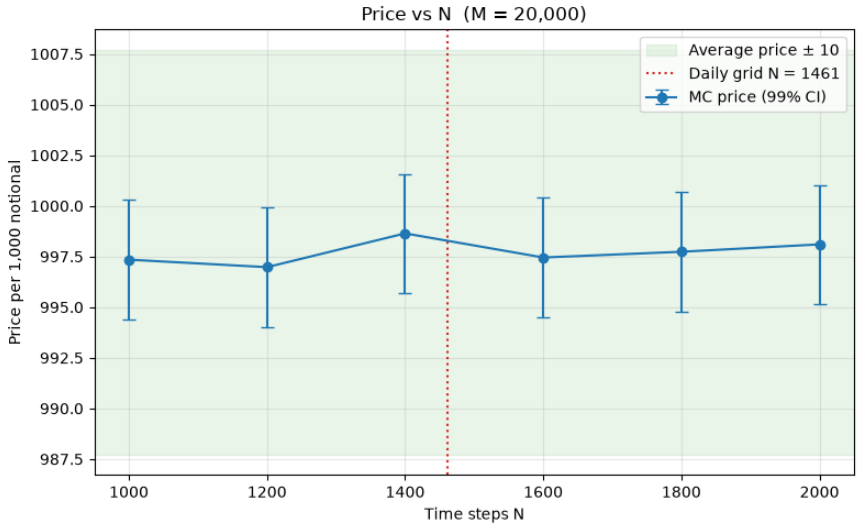


Figure 4. Price against the number of time steps, with 99% confidence intervals.

5.3 Comparison with a binomial tree

Setting $\eta = 0$ and $v_0 = \theta = \sigma^2$ turns the Heston simulator into a flat-volatility (Black–Scholes) Monte Carlo. Its prices are compared with a separately coded Cox–Ross–Rubinstein binomial tree with 14,610 steps, at five flat volatilities taken from the implied volatility surface at the note's maturity:

Moneyness	Flat vol	Flat-vol MC	99% half-width	Binomial tree
0.70	25.40%	967.05	3.35	966.29
0.80	23.89%	976.85	3.12	976.83
0.90	22.38%	986.87	2.88	986.82
0.95	21.63%	991.63	2.76	990.88
1.00	20.89%	996.67	2.63	996.12

Table 5. Flat-volatility Monte Carlo vs binomial tree.

The two methods agree to within \$0.76 at every volatility, well inside the Monte Carlo error. Since they share only the payoff definition, this confirms that the payoff, the dates, the discounting and the dividend treatment are implemented correctly.

6 The Effect of Stochastic Volatility

This section compares the stochastic volatility model and the flat-volatility model on the same note, using the flat-volatility Monte Carlo of Section 5 (same random numbers, $\eta = 0$) and the binomial tree.

6.1 Price: a flat model can match Heston, but only with the right volatility

Figure 5 shows the flat-volatility price as a function of the chosen volatility. At the at-the-money volatility (20.89%), the flat-volatility Monte Carlo gives \$996.67, \$1.89 below Heston; the difference is not statistically significant (99% CI $[-1.38,\ 5.16]$). The flat volatility that reproduces the Heston price exactly is 20.55%. But the choice is decisive: using the volatility at the 70% barrier (25.40%) prices the note at \$967.05, about \$31 too low. A flat-volatility model gives the right price only if the right volatility is picked.

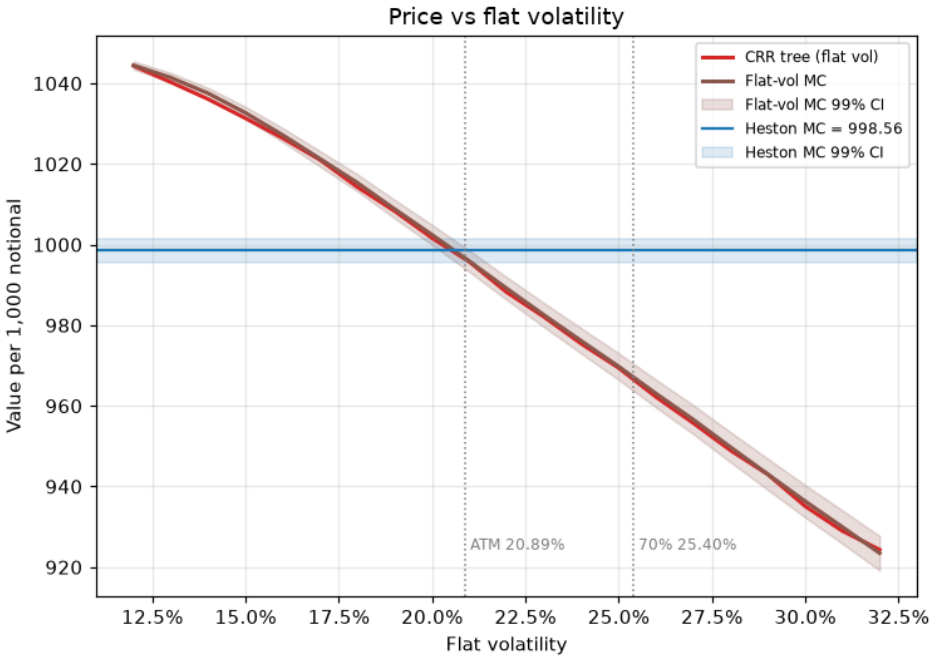


Figure 5. Price against a single flat volatility: binomial tree and flat-volatility Monte Carlo (with 99% CI), against the Heston price. Dotted lines: at-the-money and 70% moneyness volatilities.

6.2 Value: same price, different components

Even with nearly equal prices, the two models build the value differently (Tables 6 and 7).

Model	PV coupons	PV autocall principal	PV maturity principal	Total
Heston MC	101.7	817.7	79.2	998.6
Flat MC, ATM vol	118.6	751.2	126.9	996.7
Binomial tree, ATM vol	119.6	745.7	130.8	996.1

Table 6. Present value by cash flow type, per $1,000.

Model	Call at 1st date	Called by maturity	Principal loss	Coupons paid	Expected life
Heston MC	67.9%	86.4%	7.4%	5.3	1.60 yrs
Flat MC, ATM vol	52.4%	79.9%	11.1%	6.3	1.88 yrs
Binomial tree, ATM vol	51.7%	79.3%	11.2%	6.3	1.90 yrs

Table 7. Autocall and loss probabilities (risk-neutral).

Under Heston, $818 of the value comes from early redemption against $751 under flat volatility, and much less comes from coupons and from principal at maturity. The reason is the volatility term structure: Heston starts from a lower current volatility (16.8%) than the four-year flat volatility, so in the first year the index moves less and is more likely to be above its initial level on the first call date (68% vs 52%). The note is therefore expected to end about 0.3 years earlier and pay one coupon fewer.

6.3 Risk: fewer losses, but deeper ones

Heston also gives a lower probability of losing principal (7.4% vs 11.1%), but when a loss occurs it is much larger (Table 8 and Figure 6).

The flat-volatility model almost never produces a loss beyond 70% of principal, while Heston does so about seventeen times more often. The fatter left tail comes from the negative spot–volatility correlation ($\rho = -0.545$): falls in the index tend to coincide with rising volatility, as observed in equity markets, which makes large falls more likely.

	Heston MC	Flat MC, ATM vol
Probability of principal loss	7.4%	11.1%
Average loss, given a loss	55% of principal	44% of principal
Index below 30% of S_0 at maturity	1.7%	0.1%
Worst 1% of outcomes (payoff below)	$244	$432

Table 8. Severity of losses.

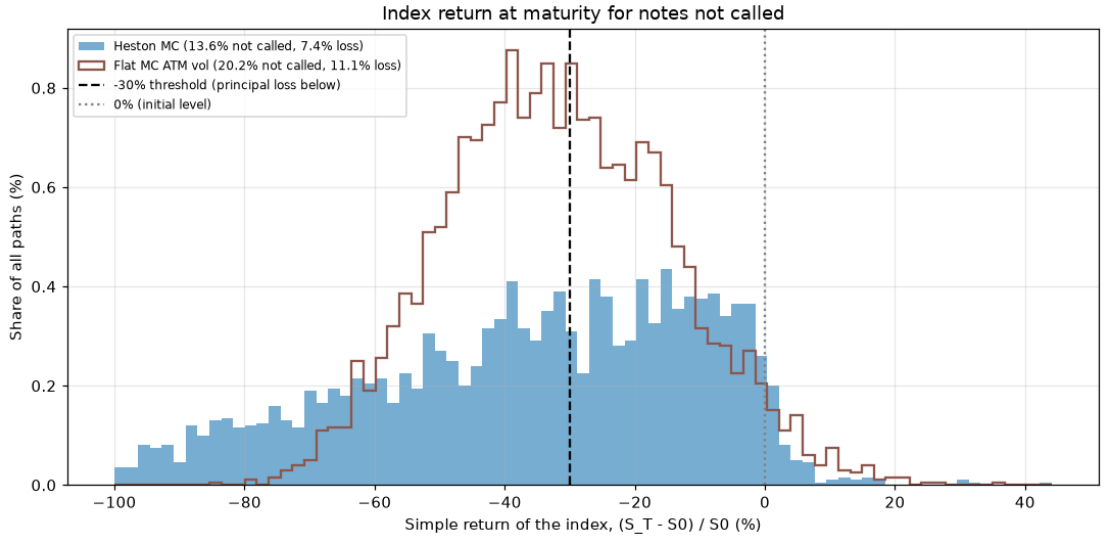


Figure 6. Index return at maturity, $(S_T - S_0)/S_0$, for notes not called, as a share of all paths: Heston (blue) vs flat-volatility Monte Carlo at the ATM volatility (brown). The area left of −30% is the probability of losing principal.

6.4 Takeaway

At the right volatility, a flat-volatility model reproduces the Heston price within simulation error, but it misstates everything behind it: it expects the note to last longer, pay more coupons and lose principal more often but less severely. Since early redemption and the size of losses determine how the note behaves and how it should be hedged, the stochastic volatility model gives a materially different, and more realistic, risk profile even when the headline price is the same.

7 Limitations and Unsolved Issues

7.1 Limitations

Table 9 lists the main simplifications and how much each one affects the result.

Limitation	**Effect on the result**	**How it was handled**
Constant r and q	The model uses one rate and one dividend yield to maturity instead of term structures. The price moves about $3.3 per 1% change in q.	Both are matched to the 4-year horizon (OIS 4.567%, OVME dividend 1.151%).
Feller condition violated	$2\kappa\theta/\eta^2 = 0.098 < 1$, so the variance spends a lot of time near zero. Euler steps then often go negative (about 45% of daily steps). Full truncation replaces v with $\max(v, 0)$, which adds a bias that depends on the step size.	The bias shrinks as the step size shrinks. Doubling the steps from $N = 1000$ to $N = 2000$ moves the price by only $1.67, less than the 99% MC half-width (about $2.9), so the bias is not visible at daily steps.
Discounting at OIS	BofA discounts at its own funding rate, so my price is above the issuer's estimated value.	I back out the implied funding spread over OIS (0.40%), which explains the gap.
Flat-vol benchmark choices	"ATM" means strike = spot rather than strike = forward. The 70% vol is extrapolated beyond the quoted strikes at long expiries.	These only affect the flat-vol benchmark, not the Heston price.

Table 9. Main limitations and how they were handled.

7.2 Unsolved Issues

1. The Feller penalty weight is not tuned: The Feller-penalized fit adds $\left[\lambda \max(0,\ \eta^2 - 2\kappa\theta)\right]^2$ to the objective with a fixed $\lambda = 10^5$. I did not search over λ, so I do not know where the best compromise between fit error and a well-behaved variance process lies, or how much the note price moves along that path.

2. The volatility surface is not from the pricing date: The calibration uses the live surface from 12-Oct-2026 because the Bloomberg historical surface did not look reliable. On the pricing date, OVME shows an ATM vol of 22.36%, against 20.89% in my surface. The note loses value as volatility rises, so my price is probably too high, but I have not measured by how much.

3. No Greeks: The report gives a price and a risk profile but no sensitivities. Hedging the note would need at least delta, vega and the sensitivity to the spot–vol correlation ρ.

A The Calibration Optimization Problem Formulation

Notation: Let $\Theta = (v_0, \kappa, \theta, \eta, \rho)$. For option i with forward F_i, strike K_i, time to expiry τ_i and type (call or put), let $C_i^H(\Theta)$ be its Heston price with parameter Θ and

$$\sigma_i^H(\Theta) = \mathrm{BS76}^{-1}\big(C_i^H(\Theta);\ F_i, K_i, \tau_i\big)$$

is its Heston implied vol, found by inverting Black-76. Let σ_i^{mkt} be the market mid IV, and give each option a weight equal to its bid–ask IV spread with a floor,

$$w_i = \max\big(0.0005,\ \sigma_i^{\mathrm{ask}} - \sigma_i^{\mathrm{bid}}\big),$$

so that the quote gets no more weight than a 0.05 vol-point spread would have.

Step 1: κ and θ from the ATM term structure. Under Heston, the expected average variance over $[0, T]$, conditional on v_0, is

$$\bar{v}(T; \kappa, \theta, v_0) := \frac{1}{T}\,\mathbb{E}\left[\int_0^T v_t\, dt \,\middle|\, v_0\right] = \theta + (v_0 - \theta)\,\frac{1 - e^{-\kappa T}}{\kappa T}.$$

With v_0 fixed at the ATM-forward variance of the 16-Oct expiry, solve

$$\min_{\kappa,\,\theta} \sum_j \left[\bar{v}(T_j; \kappa, \theta, v_0) - \sigma_{\mathrm{ATMF}}(T_j)^2\right]^2 \quad \text{s.t.} \quad 0.1 \le \kappa \le 5, \quad 10^{-4} \le \theta \le 1,$$

over the ATM-forward variance term structure.

Step 2: η and ρ from the Dec-26 and Mar-27 smiles. With (v_0, κ, θ) fixed from Step 1, solve

$$\min_{\eta,\,\rho}\ L(\Theta) \quad \text{s.t.} \quad 0.05 \le \eta \le 3, \quad -0.95 \le \rho \le 0.95,$$

for each of the three objectives below.

Step 3: all five parameters on the full surface. Starting from the Step 1–2 estimates, solve

$$\begin{aligned}\min_{\Theta}\ L(\Theta) \quad \text{s.t.} \quad & 10^{-4} \le v_0 \le 1, \quad 0.1 \le \kappa \le 5, \quad 10^{-4} \le \theta \le 1, \\ & 0.05 \le \eta \le 3, \quad -0.95 \le \rho \le 0.95,\end{aligned}$$

over 200 options across 8 expiries.

Objectives:

$$\text{Regular:} \quad L_1(\Theta) = \sum_i \left[\sigma_i^H(\Theta) - \sigma_i^{\mathrm{mkt}}\right]^2,$$

$$\text{Spread-weighted:} \quad L_2(\Theta) = \sum_i \left[\frac{\sigma_i^H(\Theta) - \sigma_i^{\mathrm{mkt}}}{w_i}\right]^2,$$

$$\text{Feller-penalized:} \quad L_3(\Theta) = L_2(\Theta) + \left[\lambda\, \max\left(0,\ \eta^2 - 2\kappa\theta\right)\right]^2, \qquad \lambda = 10^5.$$

The regular objective treats every quote equally. The weighted objective trusts tight quotes more. The Feller objective adds a penalty that is zero when $2\kappa\theta \ge \eta^2$ and grows quickly when the condition is violated. All problems are solved by bounded nonlinear least squares (`scipy.optimize.least_squares`).

Results: Table 10 reports the Step 3 solution under each objective. Errors are Heston IV minus market mid IV, in vol points.

	Regular (used)	**Spread-weighted**	**Feller-penalized**
v_0	0.0283	0.0191	0.0146
κ	3.4158	4.2030	1.8802
θ	0.0547	0.0481	0.0473
η	1.9497	1.5952	0.4216
ρ	−0.5450	−0.6037	−0.9500
Feller ratio $2\kappa\theta/\eta^2$	0.098	0.159	1.00
RMSE (vol pts)	0.19	0.28	0.36
Max abs error (vol pts)	0.61	1.01	1.24
Mean error (vol pts)	−0.00	−0.05	−0.04

Table 10. Step 3 calibration under the three objectives (8 expiries, 200 options).

The Feller-penalized fit meets the condition exactly, but its RMSE is nearly twice the regular fit's and ρ stops at its -0.95 bound.

B Heston Option Pricing

The calibration in Appendix A needs the Heston price of each listed option. I price them with the semi-closed-form solution, written in the form of Gatheral (2006). All prices are undiscounted (forward) prices, so no discount factor is needed.

Call price: For forward F, strike K, time to expiry τ and log-moneyness $x = \ln(F/K)$,

$$C^H(F,K,\tau) = F\,P_1 - K\,P_0, \qquad P^H = C^H - (F-K),$$

where the put follows from put–call parity, and P_0, P_1 are probabilities that the option finishes in the money (under two different measures):

$$P_j = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \mathrm{Re}\left[\frac{\exp\big(C_j(u,\tau) + D_j(u,\tau)\,v_0 + iux\big)}{iu}\right] du, \qquad j \in \{0,1\}.$$

Characteristic function: For $j \in \{0,1\}$,

$$\alpha_j = -\tfrac{1}{2}\big(u^2 - i(2j-1)u\big), \qquad \beta_j = \kappa - j\rho\eta - i\rho\eta u,$$
$$d_j = \sqrt{\beta_j^2 - 2\eta^2\alpha_j}, \qquad g_j = \frac{\beta_j - d_j}{\beta_j + d_j},$$

$$D_j(u,\tau) = \frac{\beta_j - d_j}{\eta^2}\cdot\frac{1 - e^{-d_j\tau}}{1 - g_j e^{-d_j\tau}},$$
$$C_j(u,\tau) = \frac{\kappa\theta}{\eta^2}\left[(\beta_j - d_j)\tau - 2\ln\frac{1 - g_j e^{-d_j\tau}}{1 - g_j}\right].$$

Numerical choices:

- **Branch of the square root:** I take the root d_j with positive real part. If this gives $|g_j| \geq 1$, I switch to $-d_j$ so that $|g_j| < 1$. This keeps the logarithm in C_j continuous and avoids blow-ups at long expiries.
- **Integration:** The integral is truncated to $u \in [10^{-7}, 400]$ and computed by the trapezoidal rule on 4,000 equally spaced points. The lower limit avoids the singularity at $u = 0$.

Implied volatility: The Heston implied vol σ^H is the volatility that makes the undiscounted Black-76 price equal the Heston price:

$$C^B(F,K,\tau,\sigma) = F\,N(d_1) - K\,N(d_2), \qquad d_{1,2} = \frac{\ln(F/K) \pm \frac{1}{2}\sigma^2\tau}{\sigma\sqrt{\tau}},$$

$$\sigma^H \text{ solves } C^B(F,K,\tau,\sigma^H) = C^H(F,K,\tau),$$

found by Brent's method on $\sigma \in [10^{-4}, 5]$ (puts use $P^B = KN(-d_2) - FN(-d_1)$). The market IVs are computed with the same Black-76 formula and the same forwards, so the model and market vols are on the same basis.

C Monte Carlo Pricing Algorithm

Notation: $P = \$1{,}000$ is the principal and $c = 0.02P$ the quarterly coupon. The coupon barrier and the loss threshold are both $B = 0.7S_0$, and the call level is S_0. The note has observation dates $t_1, \ldots, t_{16}$ with payment dates $\tau_1, \ldots, \tau_{16}$. $\mathcal{A} = \{4, \ldots, 15\}$ are the observation dates on which the note can be called. Cash flows are discounted at $r + s$, where s is the issuer funding spread ($s = 0$ for the fair value).

Algorithm 1 Heston Monte Carlo price of the autocallable note

Require: $\Theta = (v_0, \kappa, \theta, \eta, \rho)$, S_0, r, q, s, P, c, B, $\{t_k, \tau_k\}_{k=1}^{16}$, $\mathcal{A}$, T, M, N
Ensure: Price $\hat{V}$ and 99% confidence interval

$\Delta t \leftarrow T/N, \quad D_k \leftarrow e^{-(r+s)\tau_k}$ for $k = 1, \ldots, 16$
for $m = 1$ **to** M **do**
  $S \leftarrow S_0, \quad v \leftarrow v_0, \quad X_m \leftarrow 0, \quad k \leftarrow 1, \quad alive \leftarrow$ **true**
  for $j = 0$ **to** $N - 1$ **do**
    Draw independent $Z_1, Z_2 \sim N(0, 1)$
    $v^+ \leftarrow \max(v, 0)$ ▷ full truncation
    $S \leftarrow S \exp\big[(r - q - \tfrac{1}{2}v^+)\Delta t + \sqrt{v^+ \Delta t}\, Z_1\big]$
    $v \leftarrow v + \kappa(\theta - v^+)\Delta t + \eta\sqrt{v^+ \Delta t}\,\big(\rho Z_1 + \sqrt{1 - \rho^2}\, Z_2\big)$
    if $(j + 1)\Delta t = t_k$ **then** ▷ observation date
      if $S \geq B$ **then**
        $X_m \leftarrow X_m + c\, D_k$ ▷ contingent coupon
      end if
      if $k \in \mathcal{A}$ **and** $S \geq S_0$ **then**
        $X_m \leftarrow X_m + P\, D_k, \quad alive \leftarrow$ **false** ▷ autocall
        break
      end if
      $k \leftarrow k + 1$
    end if
  end for
  if *alive* **then** ▷ maturity, $S = S_T$
    if $S \geq B$ **then**
      $X_m \leftarrow X_m + P\, D_{16}$
    else
      $X_m \leftarrow X_m + P\,(S/S_0)\, D_{16}$ ▷ principal loss
    end if
  end if
end for
$\hat{V} \leftarrow \frac{1}{M}\sum_{m=1}^{M} X_m, \quad \hat{\sigma}^2 \leftarrow \frac{1}{M-1}\sum_{m=1}^{M}(X_m - \hat{V})^2$
return $\hat{V}$ and $\hat{V} \pm 2.576\,\hat{\sigma}/\sqrt{M}$

In the base case $M = 20{,}000$ and $N = 1{,}461$ (one step per calendar day), with seed 42.

D Convergence and Validation

D.1 Number of Paths

Let $X_1, \ldots, X_M$ be the discounted path payoffs from Appendix C, with sample mean $\hat{V}$ and sample standard deviation $\hat{\sigma}$. I require the price to be within $\varepsilon = \$10$ of the true value with high probability, and check this in two ways.

Chebyshev bound:

$$\Pr\big(|\hat{V} - V| \geq \varepsilon\big) \leq \frac{\hat{\sigma}^2}{M\varepsilon^2} = \frac{159.7^2}{20{,}000 \times 10^2} = 1.3\%.$$

CLT confidence interval:

$$\hat{V} \pm z_{0.995}\,\frac{\hat{\sigma}}{\sqrt{M}} = 998.56 \pm 2.576 \times 1.13 = [995.65,\ 1001.47].$$

The 99% half-width (\$2.91) is well inside the \$10 tolerance. The Chebyshev bound is loose but needs no assumption. The CLT interval is tight but assumes $\hat{V}$ is close to normal, which I check next.

D.2 Sampling Distribution of the Price

The payoff is far from normal: most paths pay about \$1,000 and a few lose a large part of the principal. To check that the average of 20,000 paths is nonetheless close to normal, I repeat the full pricing $R = 300$ times with seeds $1, \ldots, 300$.

Mean of the 300 prices	\$999.16
Std dev of the 300 prices	\$1.14
Standard error of one run ($\hat{\sigma}/\sqrt{M}$)	\$1.13
Skewness / excess kurtosis	0.03 / −0.03

Table 11. Sampling distribution of the MC price over 300 independent runs ($M = 20{,}000$).

The spread of the 300 prices (\$1.14) matches the standard error from a single run (\$1.13). Skewness and kurtosis are close to zero. The CLT interval is therefore reliable at $M = 20{,}000$.

D.3 Number of Time Steps

Full truncation adds a bias that shrinks with Δt (Section 7). I reprice with N from 1,000 to 2,000 steps over the 4-year life, keeping $M = 20{,}000$.

N	Price (\$)	99% half-width (\$)
1000	997.34	2.96
1200	996.97	2.97
1400	998.64	2.93
1600	997.45	2.97
1800	997.73	2.94
2000	998.10	2.93

Table 12. Heston price against the number of time steps.

The prices range over \$1.67 with no trend in N, which is less than the MC half-width. Any discretization bias left at daily steps ($N = 1{,}461$) is too small to see next to the simulation noise.

D.4 Independent Check Against a Binomial Tree

To test the payoff code, I set $\eta = 0$ and $v_0 = \theta = \sigma^2$ in the simulator, which turns Heston into constant-volatility GBM, and price the note with a CRR tree at the same volatility. The tree uses 14,610 steps ($10N$), where its price has stopped moving by more than about \$1 as steps increase.

Moneyness K/S_0	Vol	CRR tree (\$)	Flat MC (\$)	MC − tree (\$)
0.70	25.40%	966.29	967.05	0.76
0.80	23.89%	976.83	976.85	0.02
0.90	22.38%	986.82	986.87	0.06
0.95	21.63%	990.88	991.63	0.75
1.00	20.89%	996.12	996.67	0.55

Table 13. Note price at flat volatilities from the 4-year market surface: CRR tree vs flat-volatility MC ($M = 20{,}000$).

The two methods use different numerical schemes but agree within \$0.76 at every volatility, well inside the flat-MC half-width (about \$2.6). The simulation and the payoff logic are therefore implemented correctly, and the gap between Heston and the flat-volatility prices in Section 6 comes from the model, not from the implementation.

Reference: Gatheral, J. (2006). *The Volatility Surface: A Practitioner's Guide*. Wiley.